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Autonomous Red vs Blue Teaming: A New Frontier in Cybersecurity Risk and Reward

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respond to
security threats in a defensive manner by aiming to protect the organization's assets and ensure operational continuity. In the traditional sense, these teams have operated within rule-based frameworks. However, with the staggering rise of AI, several of these red and blue team functions are now increasingly being automated. By leveraging a combination of machine learning (ML) and reinforcement learning techniques, these AI systems can be called upon to perform penetration testing and defensive actions at unprecedented speed and scale.

For organizations, this evolution offers both promise and peril. On one hand, these autonomous teams can work independently to discover complex attack

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present across both AI systems and organizational governance best practices.

First, there are general contextual blind spots for AI systems. ART might be adept at figuring out the technical paths to solve a particular issue—such as identifying a misconfigured S3 bucket or an overly permissive identity and access management (IAM) role. However, it may lack the specific organizational business or domain context to make a fully educated assessment. An ART might identify that disabling access or shutting down a specific legacy server is the ideal way to stop lateral movement. But what the

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Many of the reinforcement learning models available today require a stable environment to learn and operate effectively. Stable environments, however, are rarely the case in complex large-scale modern enterprises. This model drift[2] can lead to divergences where the AI is optimizing for a transient architecture, leaving the actual production environment exposed.

Navigating the Regulatory and Logic

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...nd workflow orchestrations required before deploying autonomous systems at scale. The path forward involves moving from a simple vision to an actionable security posture. Improvement would require a 3-phased approach, depending on the enterprise's digital maturity:

- **Phase 1: The read-only prerequisite**—Before any action is taken by an ART or ABT, the system should focus primarily on observability parity, meaning that both human engineers and AI agents have necessary and reliable visibility into the same security data across the technology stack of the organization, including
    - **A standard baseline of harmonized data**. This unified data layer

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...umber of ...rcuit breaker is the automated control built into the identity or access management system that monitors the AI's actions and can halt them when necessary. The threshold is the limit set for the number of account revocations within the given period (X revocations in Y minutes). When this threshold is exceeded, the circuit breaker triggers, the system automatically reverts to manual mode, and a human is brought in to administer the issue.

- **Phase 3: Adversarial continuous validation**—Breach and attack simulation (BAS) should no longer be treated as a monthly report, but ...ample, ART ...ket is ...ying ... loop by ...human-led ...ational or

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autonomous failure, or low accuracy, must be integrated into the risk management practices of the organization.

- This can be achieved through the implementation of corporate AI risk registers, which can track incidents specifically reported by autonomous red or blue agents.

- **Traditional periodic audits should be augmented with audit logic.** This logic should include a stream of verifiable telemetry, as AI agents and systems change the environment in milliseconds.
- **Implement sandboxed digital twins of the organization's environment**. These act as verifiable recorders where auditors can replay autonomous battles to ensure AI reasoning remains within regulatory

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uals will not just perform the defense; they tune the machine that performs it. Organizations that embrace this transition, where AI agents are paired with human insights, are best positioned to develop a resilient cybersecurity ecosystem.

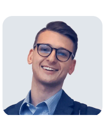
Endnotes

[1] Williams, B.; Gil, L.; "AI vs. AI: The Race Between Adversarial and Defensive Intelligence," CrowdStrike, 4 August 2025
[2] Holdsworth, J.; Stryker, C.; et al.; "What is Model Drift?," IBM
[3] Williams, Gil, "AI vs. AI"

Atish Dash

Is a skilled cybersecurity professional with over 7 years of experience in cybersecurity consulting, risk management, cloud security, and technical program management. With a diverse portfolio of certifications spanning cybersecurity, Cloud, Agile, and Program management, he has developed deep expertise in Zero-Trust Security, DevSecOps, Cloud architecture, Identity and Access management (IAM), Threat and Risk assessment, IT Service Management, and Strategic Program oversight. Atish has successfully led security initiatives for major organizations, ensuring compliance and implementing robust cybersecurity strategies. Passionate about innovation in security, he excels at problem-solving, stakeholder engagement, and driving complex projects to successful outcomes.

Additional resources



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